Automated Financial Ratio Extraction from Annual Reports

Using Large Language Models: A Consistency Study

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Abstract

We evaluate the consistency and reliability of an automated pipeline that extracts eight standardised financial ratios from corporate annual-report PDFs using Google Gemini 2.5 Flash. The pipeline applies PDF page identification via fuzzy text matching, targeted page extraction, and a two-call chain-of-thought prompting strategy. Each company is processed ten times at temperature 0.7 to measure extraction stability. Across nine multinational companies spanning technology, automotive, luxury goods, and financial services, we find that **40.3%** of all metric–company cells are consistently extracted (coefficient of variation below 5%), **48.6%** are variable, and **11.1%** are always null. Net income achieves perfect reliability (9/9 companies) while cost-to-income and interest coverage are the least reliable (0% and 11% consistent, respectively). We identify five root-cause categories for inconsistency and provide concrete recommendations for each.

Contents

1 Introduction

Extracting structured financial data from unstructured PDF annual reports is a long-standing challenge in financial natural language processing. Traditional approaches rely on hand-crafted rules or proprietary data providers. Large language models (LLMs) now offer the possibility of zero-shot extraction directly from scanned or text-based PDFs. However, LLMs are stochastic: identical prompts at non-zero temperature may yield different outputs across runs, raising concerns about reliability in downstream financial analysis.

This paper reports a controlled experiment in which the same extraction prompt is applied ten times per company to measure how consistently Gemini 2.5 Flash extracts eight financial metrics from the annual reports of nine global companies. The experiment is implemented as a four-step pipeline (Sections ??–2.5) and the results are analysed in detail in Section 4.

2 Methodology

2.1 Pipeline Overview

The pipeline consists of four sequential steps, as illustrated in Figure 1. Raw company annual-report PDFs are ingested and transformed into structured JSON records containing the extracted financial metrics and their associated consistency statistics.

Step 1: Page Identification (`table_identifier.py`)

↓

Step 2: Page Extraction (`extract_pages.py`)

↓

Step 3: Gemini Extraction $\times 10$ (`step_3_gemini_analyzer.py`)

↓

Step 4: Consistency Reporting (`step_4_overall_report.py`)

Figure 1: Four-step automated financial ratio extraction pipeline.

2.2 Step 1 — Financial Statement Page Identification

The first step identifies which pages of each annual-report PDF contain the income statement and balance sheet. This is implemented in `table_identifier.py` using the `pdfplumber` library.

2.2.1 Text-Based Table Extraction

Standard PDFs use either border-based tables (drawn with lines) or text-based layouts (columns aligned purely by whitespace). We found that text-based extraction with explicit column strategies generalises better across the diverse report formats in our dataset:

```
tables = page.extract_tables({
```

```
    "vertical_strategy":   "text",
    "horizontal_strategy": "text",
})
```

2.2.2 Fuzzy Page Matching

For each page, the top five text lines are scored against a keyword vocabulary for income statements (e.g., "consolidated statements of income", "profit and loss account") and balance sheets (e.g., "consolidated balance sheet", "statement of financial position") using the RapidFuzz `ratio` function (full-string match):

$$s_{\text{page}} = \max_{k \in K,\, l \in L_{\text{top5}}} \text{fuzz.ratio}(\text{normalise}(l),\, k) \tag{1}$$

where K is the keyword set and L_{top5} is the set of the five highest-scoring lines on the page. To handle PDFs where text is concatenated without spaces (e.g., `CONSOLIDATEDSTATEMENTOFINCOME`), the score is taken as the maximum of the original comparison and a whitespace-stripped comparison:

$$s = \max\big(\text{fuzz.ratio}(l,\, k),\ \text{fuzz.ratio}(\text{strip}(l),\, \text{strip}(k))\big) \tag{2}$$

A minimum title length of 10 characters is enforced to eliminate short false-positive matches.

2.2.3 Top-2 Page Filter

After scoring all pages, a filter retains at most the top-2 pages per statement type. A special case applies: if the top-ranked page has a strictly higher score than all others *and* all remaining pages share the same score, only the top page is retained. This avoids including auxiliary or duplicate statement pages.

2.3 Step 2 — Targeted Page Extraction

The identified income-statement and balance-sheet page numbers (physical, 1-indexed) are passed to `extract_pages.py`, which uses PyMuPDF (`fitz`) to extract only those pages into two separate PDF files per company:

- `<Company>_income_statement.pdf`
- `<Company>_balance_sheet.pdf`

Providing only the relevant pages — rather than the entire annual report — reduces token consumption and eliminates irrelevant context that could confuse the model.

2.4 Step 3 — LLM-Based Financial Ratio Extraction

2.4.1 Model and Configuration

Extraction is performed by Google Gemini 2.5 Flash via the `google-genai` SDK. Key parameters:

Parameter	Value
Model	`gemini-2.5-flash`
Temperature	0.7
Top-P	0.95
Max output tokens	16,448
Thinking mode	Disabled (`thinking_budget = 0`)
Runs per company	10

2.4.2 Two-Call Chain-of-Thought Architecture

Each run consists of two sequential Gemini API calls.

Call 1 — Chain-of-thought reasoning. The two extracted PDFs (income statement and balance sheet) are first combined into a single in-memory PDF using PyMuPDF. This combined PDF, together with the page position information, is submitted to Gemini with a structured two-step prompt:

- **Step 1:** Identify which line items are required from the income statement and balance sheet to compute each of the eight target metrics.
- **Step 2:** Read the actual values of those line items from the attached pages and calculate every metric, showing the formula and numbers used.

The model produces a free-text chain-of-thought analysis explaining its reasoning.

Call 2 — Structured JSON output. The chain-of-thought text from Call 1 is passed to a second Gemini call (without the PDF) with instructions to format the results as a structured JSON object. Each metric entry contains:

```
{
  "<metric_key>": {
    "value":   <number or null>,
    "unit":    "<e.g. millions USD, ratio, %>",
    "formula": "<formula used>",
    "inputs":  {"<line_item>": <value>, ...},
    "reason":  "<one sentence interpreting this result>"
  }, ...
}
```

If a required line item is not present in the extracted pages, the model is instructed to set `value` to `null`.

2.4.3 Target Metrics

Eight financial metrics are extracted, covering profitability, liquidity, and leverage dimensions:

Key	Metric	Formula
net_income	Net Income	Bottom-line profit
cost_to_income_ratio	Cost-to-Income	Total costs / Revenue
quick_ratio	Quick Ratio	(Current assets − Inventory) / Current liabilities
debt_to_equity_ratio	Debt / Equity	Total debt / Shareholders' equity
debt_to_assets_ratio	Debt / Assets	Total debt / Total assets
debt_to_capital_ratio	Debt / Capital	Total debt / (Debt + Equity)
debt_to_ebitda_ratio	Debt / EBITDA	Total debt / EBITDA
interest_coverage_ratio	Interest Coverage	EBIT / Interest expense

2.5 Step 4 — Consistency Measurement

After ten runs, a consistency report is generated for each company. For each metric, the ten extracted values are collected (excluding null values) and the following statistics computed:

$$\mathrm{CV} = \frac{\sigma}{|\mu|} \tag{3}$$

where μ and σ are the mean and population standard deviation of the non-null values. A metric–company cell is classified as **consistent** if and only if:

1. The null count across ten runs is zero ($n_{\text{null}} = 0$), and
2. The coefficient of variation is below the threshold $\mathrm{CV} < 0.05$ (5%).

The threshold was chosen to allow for minor rounding differences in the final decimal place while flagging genuine disagreements in the extracted values.

3 Experimental Setup

3.1 Dataset

Nine companies were selected to span a broad range of sectors, reporting currencies, and accounting frameworks:

Company	Year	Sector	Currency
Alibaba	2024	Technology / E-commerce	CNY
EM	2024	Technology	USD
GM	2023	Automotive	USD
Google	2024	Technology	USD
JPM	2024	Banking	USD
LVMH	2024	Luxury Goods	EUR
Meta	2024	Technology	USD
Tencent	2024	Technology	CNY
Volkswagen	2024	Automotive / Finance	EUR

Each company's annual report was obtained as a multi-hundred-page PDF. The pipeline extracted 1–2 income-statement pages and 1–2 balance-sheet pages per company for input to Gemini.

3.2 Evaluation Scale

Each of the 72 cells (9 companies × 8 metrics) was assigned one of three outcome labels:

Symbol	Label	Criterion
✓	Consistent	CV $< 5\%$ and $n_{\text{null}} = 0$
✗	Variable	CV $\geq 5\%$ or $n_{\text{null}} > 0$
—	All-null	All 10 runs returned `null`

4 Results

4.1 Overall Headline

Table 1 summarises the aggregate outcomes across all 72 metric–company cells.

Table 1: Overall extraction outcome across 72 cells (9 companies × 8 metrics).

Outcome	Count	Percentage	Symbol
Consistently extracted	29	40.3%	✓
Variable / noisy	35	48.6%	✗
Always null	8	11.1%	—
Total	72	100%	

4.2 Consistency Grid

Table 2 presents the full consistency grid. Company names are abbreviated: ALI = Alibaba, EM = EM, GM = GM, GOO = Google, JPM = JPM, LVM = LVMH, MET = Meta, TEN = Tencent, VW = Volkswagen. Cells are shaded: green = consistent, red = variable, grey = all-null.

Table 2: Consistency grid: ✓ = consistent (CV<5%), ✗ = variable, — = all-null.

Metric	ALI	EM	GM	GOO	JPM	LVM	MET	TEN	VW
Net Income	✓	✓	✓	✓	✓	✓	✓	✓	✓
Cost-to-Income	✗	✗	✗	✗	✗	✗	✗	✗	✗
Quick Ratio	✗	✓	✗	✓	✗	✗	✓	✗	—
Debt / Equity	✓	✓	✓	✓	✗	✗	✓	✓	—
Debt / Assets	✓	✓	✗	✗	✗	✗	✓	✓	✗
Debt / Capital	✓	✓	✗	✓	✗	✗	✓	✓	—
Debt / EBITDA	✗	✓	✗	—	✗	—	✗	—	—
Interest Coverage	✗	✗	✗	—	✗	✗	✗	✓	✗
Consistent	4	6	2	4	1	1	5	5	1
(% of 8)	50	75	25	50	13	13	63	63	13

4.3 Per-Metric Results

Table 3 reports consistency statistics aggregated across the nine companies for each metric.

Table 3: Per-metric consistency summary across nine companies.

Metric	Consistent companies	% consistent	Always-null companies	Median CV
Net Income	9	100.0%	0	0.0000
Debt / Equity	6	66.7%	1	0.0334
Debt / Capital	5	55.6%	1	0.0445
Debt / Assets	4	44.4%	0	0.0380
Quick Ratio	3	33.3%	1	0.1228
Cost-to-Income	0	0.0%	0	1.9048
Debt / EBITDA	1	11.1%	4	0.0088
Interest Coverage	1	11.1%	1	0.0683

4.3.1 Net Income (100% consistent)

Net income is the only metric achieving universal consistency. Eight of nine companies return an identical value across all ten runs (CV = 0). The sole exception, GM, shows a CV of 0.013 caused by ambiguity between consolidated net income ($10,127M) and net income attributable to stockholders ($9,840M); both values are well-defined and the variance is negligible. Net income is prominently labelled as the final line of every income statement, making it the easiest target for extraction.

4.3.2 Debt / Equity (67% consistent)

Six of nine companies pass the consistency test. Failures are structurally motivated:

- **LVMH** (CV = 0.291): Bimodal split between 0.33 and 0.59, reflecting inconsistent inclusion of operating lease obligations in the debt numerator.
- **JPM** (CV = 1.095): Bank liabilities make "debt" undefined in standard corporate terms; one run returns 9.16× (includes deposits) while others return ≈1.32× (bonds only).
- **Volkswagen**: All 10 runs null — insufficient balance sheet data in extracted pages.

4.3.3 Debt / Capital and Debt / Assets (56% and 44% consistent)

These metrics share similar failure patterns with D/E. GM additionally exhibits a unit-ambiguity failure (expressing the ratio as 44.58 instead of 0.4458), inflating CV to ≈1.9. The unit problem is mechanically identical to the cost-to-income issue and is fixable with a prompt change.

4.3.4 Quick Ratio (33% consistent)

Three companies achieve consistency (EM, Google, Meta). Failures arise from:

1. **Definition ambiguity**: Which short-term investments count as "quick assets" differs by company (Alibaba: two clusters at 1.30 and 1.79).

2. **Structural inapplicability**: JPM (a bank) lacks conventional current assets/liabilities; CV = 0.877, range 0.44–5.78.
3. **Data unavailability**: Tencent and Volkswagen return null in 9/10 and 10/10 runs respectively, indicating the balance sheet pages do not present current-asset breakdowns in a parseable format.

4.3.5 Cost-to-Income Ratio (0% consistent)

This is the only metric with *zero* consistent companies. However, the underlying extraction is largely correct — the failure is entirely a unit-expression problem. Table 4 shows that Gemini extracts the same underlying ratio value in most runs but occasionally switches between expressing it as a decimal (e.g., 0.8454) and as a percentage (e.g., 84.54). This single ambiguity inflates CV values to the range 0.80–2.73.

Table 4: Cost-to-income ratio: extracted values and CV by company. The underlying ratio is consistent; only the unit differs.

Company	Decimal form	% form	Runs as decimal	Runs as %	CV
Alibaba	0.845–0.880	84.54–88.0	7	3	1.905
EM	0.847	84.68	9	1	2.725
GM	0.946	94.59	5	5	1.195
Google	0.679	67.89	9	1	2.725
JPM	0.517	51.70	5	5	1.195
LVMH	0.769–0.777	44.74–77.7	4	4	1.232
Meta	0.578	57.82	9	1	2.725
Tencent	0.471–0.974	47.10	8	1	2.666
Volkswagen	0.817–0.944	81.68–81.69	6	4	0.802

4.3.6 Debt / EBITDA (11% consistent)

This metric suffers primarily from data unavailability. EBITDA requires depreciation and amortisation (D&A) figures, which are typically disclosed in the cash flow statement or in notes — not on the income statement or balance sheet pages provided to Gemini. Only EM (which presents D&A data on the extracted pages) achieves consistent extraction across all 10 runs (CV = 0.009). Four companies return null in every run (Google, LVMH, Tencent, Volkswagen); four others extract values in only 1–3 of 10 runs.

4.3.7 Interest Coverage (11% consistent)

Only Tencent passes the threshold (CV = 0.030). Failures fall into three sub-categories:

1. **Structural inapplicability** (JPM, GM, Volkswagen): Financial services divisions make it unclear which "interest expense" to use. GM shows a bimodal split between ≈0.76× (total financing costs) and ≈10–12× (automotive segment only), yielding CV = 0.732.
2. **Single-run outlier** (Alibaba, LVMH, EM): Nine of ten runs agree; one rogue run picks a different EBIT figure (e.g., EM returns 87.5× in one run vs. ≈63× in all others).

3. **Data unavailability** (Google): All ten runs null — Google has minimal interest expense and it is not clearly displayed on the extracted pages.

4.4 Per-Company Results

Table 5 provides detailed statistics for each company across all eight metrics.

Table 5: Per-company extraction results. Mean values are shown with the CV in parentheses. ✓ = consistent, ✗ = variable, — = all-null. Units: net income in millions (local currency); ratios are dimensionless.

Metric	ALI	EM	GM	GOO	JPM	LVM	MET	TEN	VW
Net Income (millions, local currency)									
Mean	71332	36010	10041	100118	58471	12550	62360	196467	12394
CV	0.000	0.000	0.013	0.000	0.000	0.000	0.000	0.000	0.000
	✓	✓	✓	✓	✓	✓	✓	✓	✓
Cost-to-Income Ratio									
Mean	17.94	9.23	38.40	7.40	20.99	28.09	6.30	5.23	49.36
CV	1.905	2.725	1.195	2.725	1.195	1.232	2.725	2.666	0.802
	✗	✗	✗	✗	✗	✗	✗	✗	✗
Quick Ratio									
Mean	1.656	1.065	0.726	1.660	1.869	0.636	2.822	—	—
CV	0.123	0.000	0.229	0.000	0.877	0.155	0.000	—	—
	✗	✓	✗	✓	✗	✗	✓	✗	—
Debt / Equity									
Mean	0.167	0.196	1.893	0.033	2.143	0.434	0.158	0.283	—
CV	0.049	0.011	0.001	0.032	1.095	0.291	0.004	0.033	—
	✓	✓	✓	✓	✗	✗	✓	✓	—
Debt / Assets									
Mean	0.097	0.111	9.273	0.024	2.300	0.201	0.104	0.161	0.317
CV	0.010	0.001	1.904	0.053	1.830	0.295	0.013	0.038	0.000
	✓	✓	✗	✗	✗	✗	✓	✓	✗
Debt / Capital									
Mean	0.144	0.164	13.61	0.032	11.56	0.298	0.137	0.221	—
CV	0.045	0.009	1.904	0.022	1.894	0.200	0.008	0.026	—
	✓	✓	✗	✓	✗	✗	✓	✓	—
Debt / EBITDA									
Mean	1.206	0.564	13.09	—	2.355	—	0.416	—	—

4.4.1 EM — Best Performer (6/8 consistent, 75%)

EM achieves the highest consistency score. Six of eight metrics pass the threshold, including the rarely-consistent Debt/EBITDA (CV = 0.009). The only failures are cost-to-income (universal unit ambiguity) and interest coverage (one outlier run: 87.5× vs. ≈63× in the other nine runs). EM's clean income statement layout and the availability of D&A figures on the extracted pages explain its strong performance.

4.4.2 Meta and Tencent (5/8 consistent, 63%)

Both companies benefit from straightforward balance sheets with low leverage and clearly labelled line items. Meta fails on Debt/EBITDA and Interest Coverage primarily due to sparse data (minimal interest expense not prominently displayed). Tencent's quick ratio is unreliable (9/10 null), suggesting its balance sheet format does not clearly separate current assets.

4.4.3 Alibaba and Google (4/8 consistent, 50%)

Alibaba's moderate performance reflects classification ambiguity in short-term investment items for the quick ratio and a near-borderline CV for D/Equity (0.049) and D/Capital (0.045). Google achieves perfect extraction for Net Income, Quick Ratio, D/Equity, and D/Capital but returns null for Debt/EBITDA and Interest Coverage (Google has near-zero debt and negligible interest expense).

4.4.4 GM (2/8 consistent, 25%)

GM fails on six metrics. The captive finance arm (GM Financial) introduces structural ambiguity into quick ratio, interest coverage, and leverage ratios. Additionally, GM exhibits the unit-ambiguity failure in Debt/Assets and Debt/Capital alongside cost-to-income, showing the most severe expression of this issue (50/50 decimal/percentage split).

4.4.5 JPM, LVMH, and Volkswagen (1/8 consistent, 13%)

All three bottom performers achieve only net income consistency.

- **JPM**: Standard corporate ratios are structurally incompatible with a globally systemically important bank. All seven non-income metrics fail, with CVs ranging from 0.031 (Debt/EBITDA, but 8/10 null) to 1.894 (Debt/Capital).
- **LVMH**: A luxury-goods conglomerate where significant lease obligations generate a stable bimodal split in all leverage ratios. D/Equity, D/Assets, and D/Capital each oscillate between two internally consistent but mutually exclusive debt definitions.
- **Volkswagen**: The highest null rate of any company (4 metrics always null, 3 more with 9/10 null). This is a data availability issue: VW's complex automotive + financial-services structure, combined with German IFRS reporting conventions, results in the required line items not being present on the extracted pages.

5 Root Cause Analysis

The inconsistencies observed across 35 variable cells reduce to five distinct root-cause categories, ordered by severity and ease of remediation.

5.1 Category A — Unit / Scale Ambiguity

Affected: Cost-to-income (all 9 companies), Debt/Assets and Debt/Capital for GM.

Pattern: Gemini extracts the correct underlying numerical value but expresses it inconsistently as either a decimal fraction (e.g., 0.9459) or a percentage (e.g., 94.59). At temperature 0.7, the model's output distribution covers both conventions, leading to catastrophically high CV values despite the actual information being correct.

Fix: Explicit unit specification in the prompt. For example:
"Express cost-to-income as a percentage on a 0–100 scale."
"Express all ratios as decimals in the range 0–1."

This is the single highest-impact fix: it would immediately improve cost-to-income from 0/9 to a predicted 8–9/9 consistent.

5.2 Category B — Debt Definition Ambiguity

Affected: LVMH D/E, D/A, D/Capital; JPM D/E, D/A, D/Capital.

Pattern: The balance sheet contains two or more legitimate candidates for "total debt" — typically a narrow definition (short-term borrowings + long-term debt) and a broad definition (adding lease obligations, deposits, or other financial liabilities). Gemini alternates between them depending on the chain-of-thought reasoning path taken. For LVMH, this produces clean bimodal clusters (e.g., 0.33 and 0.59 for D/E in alternating runs).

Fix: Precisely define the debt numerator in the prompt: *"Total financial debt = short-term borrowings + current portion of long-term debt + long-term debt. Exclude operating lease liabilities, trade payables, and accrued liabilities."*

5.3 Category C — Structural Inapplicability

Affected: Most metrics for JPM; Quick Ratio and Interest Coverage for GM and Volkswagen.

Pattern: Standard corporate financial ratios were developed for non-financial industrial companies. Applying them to banks (JPM) or to conglomerates with captive finance arms (GM Financial, VW Financial Services) yields results that are neither wrong nor right — different analysts would use fundamentally different denominators. The model correctly identifies multiple plausible calculations and picks one per run, producing high variance.

Fix: This requires metric redesign rather than prompt tuning. For banks: replace D/E, D/A, Quick Ratio, and Interest Coverage with institution-specific metrics (Tier-1 Capital Ratio, Net Interest Margin, Loan-to-Deposit Ratio, Return on Assets). For automotive conglomerates: specify segment isolation ("use automotive operations only, exclude financial services subsidiaries").

5.4 Category D — Data Unavailability

Affected: Debt/EBITDA (5 companies null or sparse), Interest Coverage for Google and Volkswagen, Quick Ratio for Tencent and Volkswagen.

Pattern: The required input line items do not appear on the extracted income statement and balance sheet pages. EBITDA requires depreciation and amortisation (D&A), which is typically disclosed in the cash flow statement or supplemental notes, not on the income statement. Similarly, some companies embed interest expense in notes rather than on the face of the income statement.

Fix: Extend the page extraction to include the first page of the cash flow statement (to obtain D&A) and relevant notes. This is a data engineering change, not a model change.

5.5 Category E — Single-Run Outlier

Affected: Alibaba Interest Coverage (1 outlier run), LVMH Interest Coverage (1 outlier), EM Interest Coverage (1 outlier), GM Net Income (minor).

Pattern: Nine of ten runs agree on a value; one rogue run selects a slightly different (but legitimate) line item. These cells are "almost consistent" and would pass the threshold at lower temperature.

Fix: Reduce temperature from 0.7 to 0.3 or 0.0 for production deployments. This category represents the easiest win for the five affected cells.

6 Discussion and Recommendations

6.1 Metric Reliability Ranking

Table 6 ranks the eight metrics by their production readiness based on observed consistency rates and root-cause severity.

Table 6: Metric reliability ranking and recommended actions.

Rank	Metric	Consistent	Action
1	Net Income	9/9	Production-ready as-is
2	Debt / Equity	6/9	Specify debt definition; skip JPM
3	Debt / Capital	5/9	Specify unit + debt definition
4	Debt / Assets	4/9	Specify unit + debt definition
5	Quick Ratio	3/9	Specify quick-asset components
6	Debt / EBITDA	1/9	Add cash flow statement pages
7	Interest Coverage	1/9	Add interest expense pages; skip banks
8	Cost-to-Income	0/9	Add unit instruction (highest ROI fix)

6.2 Company Suitability for Automated Extraction

Technology companies with simple, straightforward financial statements (EM, Meta, Tencent, Google, Alibaba) are well-suited to the current pipeline. Conglomerates with complex capital structures (LVMH, Volkswagen) require additional prompt engineering. Financial institutions (JPM) require fundamentally different metrics.

6.3 Prioritised Improvement Roadmap

P1 — Prompt: unit specification (High impact, zero code change)
Add explicit unit instructions for cost-to-income, Debt/Assets, and Debt/Capital. Expected to fix cost-to-income for all 9 companies and Debt/Assets/Capital for GM. Estimated improvement: +13 consistent cells.

P2 — Prompt: debt definition (High impact, zero code change)
Precisely define the debt numerator. Expected to stabilise D/E, D/A, D/Capital for LVMH and partially for JPM. Estimated improvement: +3–6 cells.

P3 — Data: add cash flow pages (Medium effort)
Include the first page of the operating cash flow section. Required for reliable Debt/EBITDA extraction. Expected improvement: +4–5 cells.

P4 — Metric redesign for banks (High effort)
Replace all seven non-income metrics for JPM with bank-specific KPIs. Expected improvement: +6 cells for JPM alone.

P5 — Temperature reduction (Zero effort)
Reduce temperature from 0.7 to 0.3. Eliminates Category E single-run outliers. Estimated improvement: +3–5 cells.

7 Conclusion

This study evaluates the reproducibility of LLM-based financial ratio extraction across nine global companies and eight metrics over ten independent runs at temperature 0.7. The key findings are:

1. **Net income extraction is production-ready** across all sectors (100% consistent).
2. **Leverage ratios are moderately reliable** for non-financial companies: D/Equity achieves 67%, D/Capital 56%, D/Assets 44%.
3. **Cost-to-income fails universally** due to a single, fixable unit-ambiguity problem. The underlying extracted value is correct; only its scale fluctuates.
4. **Debt/EBITDA and Interest Coverage are data-limited**, not model-limited. Including cash flow statement pages would substantially improve both metrics.
5. **Sector matters**: Technology companies (EM, Meta, Google, Tencent) significantly outperform financial institutions (JPM) and complex automotive conglomerates (Volkswagen), which require sector-specific metric definitions.
6. **Three targeted prompt changes** — unit specification, debt definition, and quick-asset enumeration — are predicted to raise overall consistency from 40% to

approximately 60–70% without any code modifications.

The two-call chain-of-thought architecture proves effective at guiding the model through multi-step financial calculations, and the pipeline's modular design allows each identified weakness to be addressed independently. Future work should evaluate the effect of lower temperatures on consistency, extend page coverage to include notes and cash flow statements, and develop sector-specific extraction profiles for financial institutions.

A Raw Extracted Values per Company

Tables 7–15 list the complete set of extracted values across all ten runs for each company.

Table 7: Alibaba 2024 — all extracted values (10 runs).

Metric	**Values**	**Mean**	**CV**	
Net Income	71332 × 10	71332	0.000	✓
Cost-to-Income	0.845, 0.845, 0.880, 0.880, 88.0, 0.845, 84.54, 0.845, 0.845, 0.880	17.94	1.905	✗
Quick Ratio	1.786, 1.303, 1.786, 1.786, 1.790, 1.446, 1.786, 1.303, 1.786, 1.790	1.656	0.123	✗
Debt / Equity	0.173, 0.155, 0.173, 0.173, 0.173, 0.155, 0.173, 0.155, 0.173, 0.170	0.167	0.049	✓
Debt / Assets	0.097, 0.097, 0.097, 0.097, 0.097, 0.097, 0.097, 0.097, 0.097, 0.100	0.097	0.010	✓
Debt / Capital	0.148, 0.134, 0.148, 0.148, 0.148, 0.134, 0.148, 0.134, 0.148, 0.150	0.144	0.045	✓
Debt / EBITDA	1.174, 1.174, 1.270, null × 7	1.206	0.038	✗
Interest Coverage	16.98, 14.26, 14.26, 14.26, 14.26, 14.26, 14.26, 14.26, 14.26, null	14.56	0.059	✗

Table 8: EM 2024 — all extracted values (10 runs).

Metric	**Values**	**Mean**	**CV**	
Net Income	36010 × 10	36010	0.000	✓
Cost-to-Income	0.847 × 9, 84.68	9.23	2.725	✗
Quick Ratio	1.065 × 9, 1.065	1.065	0.000	✓
Debt / Equity	0.196, 0.203, 0.196, 0.196, 0.196, 0.196, 0.196, 0.196, 0.196, 0.196	0.196	0.011	✓
Debt / Assets	0.111 × 9, 0.111	0.111	0.001	✓
Debt / Capital	0.164, 0.169, 0.164, 0.164, 0.164, 0.164, 0.164, 0.164, 0.164, 0.164	0.164	0.009	✓
Debt / EBITDA	0.570, 0.560, 0.560, 0.570, 0.560, 0.570, 0.560, 0.570, 0.560, 0.560	0.564	0.009	✓
Interest Coverage	63.17, 63.17, 63.17, 61.59, 63.17, 62.17, 87.48, 63.17, 63.17, 62.17	65.24	0.114	✗

Table 9: GM 2023 — all extracted values (10 runs).

Metric	Values	Mean	CV	
Net Income	10127 × 7, 9840 × 3	10041	0.013	✓
Cost-to-Income	94.59, 0.946, 94.59, 0.946, 94.59, 94.59, 0.946, 0.946, 0.946, 0.946	38.40	1.195	✗
Quick Ratio	0.830, 0.825, 0.410, 0.825, 0.825, 0.830, 0.825, 0.651, 0.825, 0.411	0.726	0.229	✗
Debt / Equity	1.890, 1.894, 1.890, 1.894, 1.894, 1.890, 1.894, 1.894, 1.894, 1.894	1.893	0.001	✓
Debt / Assets	0.450, 0.446, 44.58, 0.446, 0.446, 44.58, 0.446, 0.446, 0.446, 0.446	9.273	1.904	✗
Debt / Capital	0.650, 0.654, 65.44, 0.654, 0.654, 65.44, 0.654, 0.654, 0.654, 0.654	13.61	1.904	✗
Debt / EBITDA	13.09, null × 9	13.09	0.000	✗
Interest Coverage	0.760, 0.757, 10.21, 0.757, 11.42, 10.21, 0.757, 10.21, 12.42, 10.21	6.770	0.732	✗

Table 10: Google 2024 — all extracted values (10 runs).

Metric	Values	Mean	CV	
Net Income	100118 × 10	100118	0.000	✓
Cost-to-Income	0.679 × 9, 67.89	7.40	2.725	✗
Quick Ratio	1.661, 1.661, 1.661, 1.661, 1.660, 1.661, 1.661, 1.660, 1.661, 1.660	1.660	0.000	✓
Debt / Equity	0.034 × 9, 0.030	0.033	0.032	✓
Debt / Assets	0.024 × 9, 0.020	0.024	0.053	✗
Debt / Capital	0.032 × 9, 0.030	0.032	0.022	✓
Debt / EBITDA	null × 10	—	—	—
Interest Coverage	null × 10	—	—	—

Table 11: JPM 2024 — all extracted values (10 runs).

Metric	Values	Mean	CV	
Net Income	58471 × 10	58471	0.000	✓
Cost-to-Income	0.517, 0.517, 51.70, 0.517, 51.70, 0.517, 0.517, 51.69, 51.69, 0.517	20.99	1.195	✗
Quick Ratio	0.471, 3.080, 0.580, 1.970, 5.780, 1.970, 1.970, 0.555, 0.443, null	1.869	0.877	✗
Debt / Equity	1.318, 1.320, 1.320, 1.320, 1.160, 9.160, 1.318, 1.320, 1.877, 1.318	2.143	1.095	✗
Debt / Assets	0.114, 0.110, 0.110, 0.114, 10.03, 0.789, 0.114, 11.35, 0.162, 0.114	2.300	1.830	✗
Debt / Capital	0.569, 0.570, 0.570, 0.569, 53.80, 0.902, 0.569, 56.85, 0.652, 0.569	11.56	1.894	✗
Debt / EBITDA	2.280, 2.430, null × 8	2.36	0.032	✗
Interest Coverage	1.741, 0.740, 1.740, 0.740, 1.850, 0.741, 1.850, 1.741, 1.741, null	1.432	0.343	✗

Table 12: LVMH 2024 — all extracted values (10 runs).

Metric	Values	Mean	CV	
Net Income	12550 × 10	12550	0.000	✓
Cost-to-Income	44.74, 0.777, 0.777, 77.70, 76.92, 0.777, 0.777, 0.777, 0.769, 76.92	28.09	1.232	✗
Quick Ratio	0.680, 0.677, 0.706, 0.677, 0.680, 0.455, 0.677, 0.677, 0.427, 0.706	0.636	0.155	✗
Debt / Equity	0.330, 0.589, 0.331, 0.589, 0.330, 0.589, 0.331, 0.331, 0.589, 0.331	0.434	0.291	✗
Debt / Assets	0.150, 0.273, 0.154, 0.273, 0.150, 0.273, 0.154, 0.154, 0.273, 0.154	0.201	0.295	✗
Debt / Capital	0.250, 0.371, 0.249, 0.371, 0.250, 0.371, 0.249, 0.249, 0.371, 0.249	0.298	0.200	✗
Debt / EBITDA	null × 10	—	—	—
Interest Coverage	19.86 × 9, 23.87	20.26	0.059	✗

Table 13: Meta 2024 — all extracted values (10 runs).

Metric	Values	Mean	CV	
Net Income	62360 × 10	62360	0.000	✓
Cost-to-Income	0.578 × 9, 57.82	6.30	2.725	✗
Quick Ratio	2.822, 2.822, 2.822, 2.822, 2.820, 2.822, 2.822, 2.820, 2.820, 2.822	2.822	0.000	✓
Debt / Equity	0.158 × 9, 0.160	0.158	0.004	✓
Debt / Assets	0.104 × 9, 0.100	0.104	0.013	✓
Debt / Capital	0.136 × 9, 0.140	0.137	0.008	✓
Debt / EBITDA	0.416, null × 9	0.416	0.000	✗
Interest Coverage	55.08, 55.08, 54.08, null × 7	54.74	0.009	✗

Table 14: Tencent 2024 — all extracted values (10 runs).

Metric	Values	Mean	CV	
Net Income	196467 × 10	196467	0.000	✓
Cost-to-Income	0.974, 0.471, 0.471, 0.471, 47.10, 0.471, 0.471, 0.471, 0.471, 0.974	5.23	2.666	✗
Quick Ratio	3.793, null × 9	3.793	—	✗
Debt / Equity	0.273, 0.299, 0.285, 0.263, 0.280, 0.285, 0.285, 0.285, 0.290, 0.290	0.283	0.033	✓
Debt / Assets	0.161, 0.164, 0.156, 0.156, 0.160, 0.156, 0.156, 0.156, 0.172, 0.172	0.161	0.038	✓
Debt / Capital	0.214, 0.230, 0.222, 0.208, 0.220, 0.222, 0.222, 0.222, 0.225, 0.225	0.221	0.026	✓
Debt / EBITDA	null × 10	—	—	—
Interest Coverage	17.37, 17.37, 18.70, 17.37, 17.37, 17.37, 17.37, 17.37, 18.70, 17.37	17.64	0.030	✓

Table 15: Volkswagen 2024 — all extracted values (10 runs).

Metric	Values	Mean	CV	
Net Income	12394 × 10	12394	0.000	✓
Cost-to-Income	81.69, 81.68, 81.68, 81.68, 0.944, 81.68, 0.817, 81.68, 0.817, 0.944	49.36	0.802	✗
Quick Ratio	null × 10	—	—	—
Debt / Equity	null × 10	—	—	—
Debt / Assets	0.317, null × 9	0.317	0.000	✗
Debt / Capital	null × 10	—	—	—
Debt / EBITDA	null × 10	—	—	—
Interest Coverage	6.23, 5.53, 5.17, 5.53, 5.88, 5.88, 5.88, 5.53, 6.58, 6.23	5.84	0.068	✗